THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A chemical change always produces: 1

(a) one or more new substances **(b)** only a change of colour
(c) only a change of state **(d)** no new substance

2. $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ is a: 1

(a) decomposition reaction **(b)** combination reaction
(c) double displacement reaction **(d)** displacement reaction

3. The colour of copper sulphate solution is: 1

(a) colourless **(b)** yellow **(c)** green **(d)** blue

4. Substances on the left side of a chemical equation are called: 1

(a) catalysts **(b)** products **(c)** precipitates **(d)** reactants

5. $2\,H_2O \longrightarrow 2\,H_2 + O_2$ is a: 1

(a) decomposition reaction **(b)** combination reaction
(c) displacement reaction **(d)** neutralisation reaction

6. The colour of barium sulphate precipitate is: 1

(a) black **(b)** blue **(c)** white **(d)** yellow

7. The chemical formula of quick lime is: 1

(a) $CaCl_2$ **(b)** CaO **(c)** $CaCO_3$ **(d)** $Ca(OH)_2$

8. $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ is a: 1

(a) double displacement reaction **(b)** decomposition reaction
(c) combination reaction **(d)** displacement reaction

9. The colour of lead iodide precipitate is: 1

(a) white **(b)** blue **(c)** red **(d)** yellow

10. The chemical formula of slaked lime is: 1

(a) $CaCO_3$ **(b)** $Ca(OH)_2$ **(c)** CaO **(d)** $CaSO_4$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the law of conservation of mass. 2

12. Define a decomposition reaction and give an example. 2

13. What is corrosion? Name the product formed on iron. 2

14. Write the word equation for burning of magnesium in oxygen. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance the equation $\mathrm{Na} + \mathrm{Cl_2} \longrightarrow \mathrm{NaCl}$. 3

16. Name the type of reaction: $\mathrm{CaO} + \mathrm{H_2O} \longrightarrow \mathrm{Ca(OH)_2}$. Give a reason and say whether it is exothermic or endothermic. 3

17. An iron nail is kept in copper sulphate solution. State three observations. 3

OR

Balance the equation $\mathrm{N_2} + \mathrm{H_2} \longrightarrow \mathrm{NH_3}$. 3

18. Name the type of reaction: $\mathrm{CaCO_3} \longrightarrow \mathrm{CaO} + \mathrm{CO_2}$. Give a reason. 3

OR

Magnesium ribbon is burnt in air. State three observations. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A magnesium ribbon is burnt in air. State the observation, write the balanced equation, name the type of reaction and the product. Why is the ribbon cleaned first? 5

OR

What is corrosion? State the conditions necessary for rusting, name rust and give three ways to prevent it. 5

20. Lead nitrate is heated in a test tube. Write the observation, equation and type of reaction. Name the brown gas. 5

OR

What is rancidity? Why does it occur and how can it be prevented? 5

END OF QUESTION PAPER — **Total: 40 Marks**

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